

Doubly robust estimation under a possibly misspecified marginal structural Cox model

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SUMMARY

In this article we consider the marginal structural Cox model, which has been widely used to analyse observational studies with survival outcomes. The standard inverse probability weighting method under the model hinges on a propensity score model for the treatment assignment and a censoring model that incorporates both the treatment and the covariates. In such settings model misspecification can often occur, and the Cox regression model's non-collapsibility has historically posed challenges when striving to guard against model misspecification through augmentation. We introduce a novel joint augmentation to the martingale-based full-data estimating functions and develop rate double robustness, which allows the use of machine learning and nonparametric methods to overcome the challenges of non-collapsibility. We closely examine its theoretical properties to guarantee root- n inference for the estimand. The estimator extends naturally to estimating a time-average treatment effect when the proportional hazards assumption fails, and we show that it satisfies both the assumption-lean and the well-specification criteria in the context of a causal estimand for censoring survival data; that is, it is a functional of the potential outcome distributions only and does not depend on the treatment assignment mechanism, the covariate distribution or the censoring mechanism. The martingale-based augmentation approach is also applicable to many semiparametric failure time models. Finally, its application to a dataset provides insights into the impact of mid-life alcohol consumption on mortality in later life.

Some key words: Augmented inverse probability weighting; Causal hazard ratio; Causal inference; Machine learning; Time-averaged treatment effect.

1. INTRODUCTION

1.1. *Background*

The marginal structural Cox model (Hernán et al., 2001; Hernán & Robins, 2020) has been widely used in observational studies with survival outcomes to estimate the causal hazard ratio; see, for example, Cole et al. (2003), Feldman et al. (2004), Sterne et al. (2005), Hernán et al. (2006) and Buchanan et al. (2014), among many works. While the interpretation of the hazard function for causal inference has recently been under debate (see Prentice & Aragaki, 2022, Ying & Xu, 2023, and references therein), the Cox model formulation continues to be widely used and can be easily adapted to derive more commonly agreed-upon interpretable quantities, such as survival probabilities.

Common approaches to estimating parameters under the marginal structural Cox model, i.e., the causal estimands, have been inverse probability of treatment weighting (IPTW) and inverse probability of censoring weighting (IPCW) (Hernán et al., 2001). Both weighting schemes require estimation of quantities related to the conditional distribution of treatment assignment and the conditional distribution of censoring given covariates. Parametric or semiparametric models for these conditional distributions are often subject to misspecification, leading to inconsistent estimators of the estimands. More flexible approaches such as machine learning or nonparametric methods, on the other hand, give rise to estimators that converge to the true estimands at slower than $\sqrt{n}$ rates.

To overcome the aforementioned drawbacks of inverse probability weighting (IPW) approaches, augmented inverse probability weighting (AIPW) methods have been developed (Robins et al., 1995; Scharfstein et al., 1999; Robins, 2000; Robins et al., 2000a,b; Robins & Rotnitzky, 2001; van der Laan & Robins, 2003; Bang & Robins, 2005; Tsiatis, 2006). These methods often exhibit so-called doubly robust properties, to be elaborated on in more detail later. Robins (1998) developed a set of semiparametric estimators for marginal structural model parameters, emphasizing efficiency and not allowing for possible misspecification of the propensity score. A main challenge in developing doubly robust estimators under the marginal structural Cox model is the non-collapsibility of the Cox regression model (Martinussen & Vansteelandt, 2013), i.e., the Cox model formulation including the proportional hazards assumption typically no longer holds when a covariate is integrated out from the model, an issue that has been well documented since the 1980s; see Lancaster & Nickell (1980), Gail et al. (1984), Ford et al. (1995) and the 1996 PhD thesis by R. Xu of the University of California, San Diego. This gives rise to the difficulty of specifying a parametric or semiparametric conditional survival outcome model that is needed in a doubly robust approach and is at the same time compatible with the marginal structural Cox model which defines the causal estimand (Tchetgen Tchetgen & Robins, 2012). Despite previous efforts, the problem of doubly robust estimation under the marginal structural Cox model has not been satisfactorily solved.

When using the Cox proportional hazards model, a major concern is violation of the proportional hazards assumption. This also applies to the marginal structural Cox model. In such cases, the partial likelihood estimator is known to be affected by the nuisance censoring distribution, even in the absence of confounding bias; see Xu & O'Quigley (2000) and the 1996 PhD thesis of R. Xu. While the doubly robust property helps to guard against possible misspecification of the so-called working models, little work has been done in the causal inference literature to investigate what happens when the model used to define the causal estimand is misspecified.

1.2. Overview of the paper

In this article we derive an AIPW estimator under the marginal structural Cox model. A novel aspect of our approach is the joint augmentation of the estimating functions for both the log hazard ratio and the nuisance baseline hazard function under the Cox model. Unlike previous attempts using the partial likelihood under the Cox model, this joint augmentation makes use of the independent and identically distributed martingale constructs and can be more broadly applied to semiparametric models for failure time data. The augmentation leads to working models for the treatment assignment given the covariates, i.e., the propensity score, the failure time and the censoring time, given the treatment and the covariates. To circumvent the non-collapsibility problem described above and to specify a conditional failure time model that is compatible with the original marginal structural Cox model, data-adaptive machine learning or nonparametric methods are needed. We show that by using the above augmentation together with cross-fitting, the resulting estimator has a rate doubly robust property which allows $\sqrt{n}$ inference in the presence of a slower than $\sqrt{n}$ convergence rate of the working models (Rotnitzky et al., 2021; Hou et al., 2023).

Also new in this work is the consideration of a possibly misspecified marginal structural Cox model. In place of the proportional hazards assumption on the distributions of the two potential failure time outcomes, we consider a general time-varying log hazard ratio. We show that the proposed AIPW estimator converges to a well-defined and well-specified time-averaged treatment effect under the potential outcomes framework. We establish rate double robustness under this general time-varying log hazard ratio, which contains the marginal structural Cox model as a special case. Proofs of all the theoretical results are given in the [Supplementary Material](#).

1.3. Related work

For survival outcomes, AIPW approaches have been studied outside the Cox model. Rotnitzky & Robins (2005) introduced an augmented IPCW method tailored to censored survival data. Zhang & Schaubel (2012a), Bai et al. (2017) and Sjölander & Vansteelandt (2017) produced doubly robust estimators for a contrast between the expected transformed potential failure times. Yang et al. (2020) designed a doubly robust estimator for structural accelerated failure time models. Both Petersen et al. (2014) and Zheng et al. (2016) derived targeted maximum likelihood estimators with doubly robust properties by discretizing time, and framing the failure time as a binary outcome, and Rytgaard & van der Laan (2024) extended these to continuous time with possible competing risks, focusing on cumulative quantities such as the survival probabilities. Within the additive hazards model, Dukes et al. (2019) and Hou et al. (2023) presented doubly robust estimators for hazard differences across low and high dimensions, and Rava & Xu (2023) extended these to competing risks.

Robins (1998) first considered various marginal structural models including the marginal structural Cox model, proposed the IPTW approach for estimation, and performed efficiency calculation using missing data theory. Tchetgen Tchetgen & Robins (2012) applied these results to a class of estimators under the marginal structural Cox model and developed a doubly robust quantity using the approach from van der Laan & Robins (2003), which is the same as the one used here. We defer the details of the contrast with our development to the next section. Tchetgen Tchetgen & Robins (2012) then focused on finding a (semi)parametric model for the conditional distribution of the failure time given the treatment and covariate, so that it would be compatible with the original marginal structural Cox model, but without resorting to nonparametric or machine learning methods like the

ones considered in this paper. Their limited solution to the non-collapsibility problem within (semi)parametric working models has not resulted in any feasible implementation for practical applications. On the other hand, our novel joint augmentation with independent and identically distributed martingale constructs, not only bypasses the non-collapsibility issue under the marginal structural Cox model, but can be applied more broadly to other survival models that define relevant causal estimands.

Outside the causal inference context, another significant application of IPCW has emerged in situations where there is violation of the proportional hazards assumption. Several studies, including [Xu & O’Quigley \(2000\)](#), [Boyd et al. \(2012\)](#), [Hattori & Henmi \(2012\)](#), [Nguyen & Gillen \(2017\)](#), [Nuño & Gillen \(2021\)](#) and the 1996 PhD thesis of R. Xu, have worked on correcting the bias caused by a nuisance censoring mechanism using IPCW. However, not all these works explicitly use the term IPCW; some opt for conditional survival distribution increments as weights, but mathematically these align with the inverse probability of censoring weights. Among these works, [Xu & O’Quigley \(2000\)](#) and the PhD thesis of R. Xu assumed the censoring distribution to be independent of the regressors in the model. [Boyd et al. \(2012\)](#) allowed the censoring distribution to be different between the treatment groups, but otherwise independent of the covariates. [Nguyen & Gillen \(2017\)](#) allowed the censoring distribution to depend on the covariates and introduced a survival tree method to estimate the conditional censoring distribution given the covariates.

Informative censoring has recently received much attention in applications such as clinical oncology ([Campigotto & Weller, 2014](#); [Templeton et al., 2020](#); [Olivier et al., 2021](#)). Meanwhile, efforts have been made in the statistical community to select covariates to account for censoring ([Van Lancker et al., 2021](#)), find transformations that render the relevant model and parameters identifiable ([Deresá & Van Keilegom, 2021](#)) and apply copula-type approaches ([Czado & Van Keilegom, 2023](#)).

Finally, there has been discussion in the literature about model robustness and assumption-lean estimation ([Buja et al., 2019a,c](#), with discussion), in the sense that models are approximations ([Box, 1979](#)) to perhaps much more complex reality. The emphasis here is not on the possibly wrong working models for any nuisance parameters, but on the model that defines the estimand itself. As pointed out by [Whitney et al. \(2019\)](#), the examples explicitly considered in [Buja et al. \(2019a,c\)](#) were entirely parametric, while the situation with certain semiparametric models, such as asymptotic linearity of the estimator, can be quite different. [Whitney et al. \(2019\)](#) used the Cox model as an illustration, and additional work has been done on specific types of assumption-lean Cox regression ([Vansteelandt et al., 2024](#)). The present work also contributes to finding an assumption-lean solution to a well-specified estimand in that context. We will discuss this aspect further after describing our approach.

2. DOUBLY ROBUST SCORE

2.1. Marginal structural Cox model

Let A be a binary treatment. Define $T(0)$ and $T(1)$ as the potential failure times for a subject under $a = 0$ and 1 , respectively. Let $\lambda_{T(a)}(t)$ denote the hazard function of the potential failure time $T(a)$ for $a \in \{0, 1\}$. The marginal structural Cox model ([Hernán et al., 2001](#)) for the potential outcomes posits that

$$\lambda_{T(a)}(t) = \lambda_0(t) \exp(\beta a), \quad (1)$$

where $\lambda_0(t)$ represents an unknown baseline hazard function and β serves as the causal log hazard ratio, delineating the contrast between the potential failure time outcome distributions when $a = 1$ and $a = 0$. The potential failure time $T(a)$ could be right-censored by $C(a)$. Define $\Delta(a) = I\{T(a) \leq C(a)\}$, where $I(\cdot)$ is the indicator function, and $X(a) = \min\{T(a), C(a)\}$. We use T, C, X and Δ to indicate the observed counterparts once the treatment is received. Let Z denote a vector of p -dimensional covariates.

We adopt the standard causal inference assumptions (Hernán & Robins, 2020) and use $\perp$ to indicate statistical independence. We impose the stable unit treatment value assumption, meaning that a subject's potential outcomes are independent of the treatment assignment of other subjects and that there are no hidden versions of the treatments. Although model (1) does not incorporate the covariates Z , we adopt an informative censoring assumption that allows the censoring time to depend on Z .

Assumption 1.

- (i) Consistency: $T = AT(1) + (1 - A)T(0)$ and $C = AC(1) + (1 - A)C(0)$.
- (ii) Exchangeability: $(T(a), C(a)) \perp A \mid Z$ for $a = 0, 1$.
- (iii) Positivity: there exists $0 < \epsilon < 1$ such that $\epsilon < \text{pr}(A = 1 \mid Z = z) < 1 - \epsilon$, $\text{pr}(C > \tau \mid A = a, Z = z) > \epsilon$ and $\text{pr}(T > \tau \mid A = a, Z = z) > \epsilon$ for all values of a and z , where τ is a maximum follow-up time.
- (iv) Informative censoring: $T(a) \perp C(a) \mid Z$ for $a = 0, 1$.

In the following we start with the full data, which include both potential outcomes for $a = 0, 1$ and do not involve censoring. We create our full-data estimating functions for both β and Λ_0 based on the full-data martingale as per model (1) and its increments. This martingale-based joint augmentation approach, as we will show later, can be more broadly applied to other semiparametric models for failure time data. We then apply IPW, which leads to the identification of the causal estimands using the observed data only. Finally, we develop a novel joint augmentation approach that gives the desired doubly robust property.

2.2. Full-data score

When we consider both the counterfactual outcome and the censoring as missing data, the full data, using the notion in Tsiatis (2006), is $(T(0), T(1), Z)$. From this we can define for $a = 0, 1$ the full-data counting process $N_T^a(t) = I\{T(a) \leq t\}$ and the full-data at-risk process $Y_T^a(t) = I\{T(a) \geq t\}$. It can be shown that

$$M_T^a(t; \beta, \Lambda) = N_T^a(t) - \int_0^t Y_T^a(u) \exp(\beta a) d\Lambda(u)$$

is a full-data martingale with respect to the filtration $\mathcal{F}_t^a = \{N_T^a(u), Y_T^a(u^+) : 0 \leq u \leq t\}$ under model (1), where $\Lambda(t) = \int_0^t \lambda_0(u) du$ (Fleming & Harrington, 1991). We start by constructing a full-data score function, i.e., an estimating function we would use if we were able to observe a single copy of the full data. Using the martingale property, we define the full-data scores for $\Lambda(t)$ and β as

$$D_1^f(t; \beta, \Lambda) = \sum_{a=0}^1 dM_T^a(t; \beta, \Lambda), \quad (2)$$

$$D_2^f(\beta, \Lambda) = \sum_{a=0}^1 \int_0^\tau a \, dM_T^a(t; \beta, \Lambda). \quad (3)$$

Here, $D_1^f(\beta, \Lambda, t)$ is a martingale difference function that is often used in survival analysis; see, for example, [Lu & Ying \(2004\)](#). For each t , the true values of β and $\Lambda(t)$ satisfy $E\{D_1^f(\beta, \Lambda, t)\} = 0$ and $E\{D_2^f(\beta, \Lambda)\} = 0$. In addition, it can be readily verified that for a random sample of size n , these would give the well-known Breslow's estimate of $\Lambda(t)$, as well as the partial likelihood score for β .

At the suggestion of a reviewer, here we contrast our approach with the development in [Tchetgen Tchetgen & Robins \(2012\)](#). Instead of the partial likelihood score, which is an integrated difference over time between the observed treatment variable A and its expectation in the risk set under the Cox model, [Tchetgen Tchetgen & Robins \(2012\)](#) consider a more general quantity $g(A, t)$ and its expectation in the risk set under the Cox model. This is known to give a consistent estimate of the treatment effect when IPTW is applied. The D_2^f in (3) may also be redefined using $g(a, t)$ in place of a in the right-hand side, if we so choose. It is known that under randomized treatment assignment, i.e., the classic Cox model with A as the regressor, the choice of $g(a, t) = a$ gives the partial likelihood score and is semi-parametrically efficient. Under violation of the Cox model assumption, different choices of $g(a, t)$ will likely give rise to different estimands, but to the best of our knowledge this has not been systematically examined in the literature. [Tchetgen Tchetgen & Robins \(2012\)](#) chose a $g(a, t)$ that gives a closed-form estimator of β . In order to achieve double robustness to safeguard against misspecification of the propensity score model needed in IPTW, they then applied augmentation to a certain key quantity in the closed-form expression of their estimator. Here we instead use a novel approach that consists of joint augmentation of the two IPW scores (4) and (5). This approach plays a key role in rigorously establishing rate double robustness of our estimator, which allows the use of nonparametric or machine learning methods to bypass the non-collapsibility issue under the marginal structural Cox model. The joint augmentation also gives an immediate estimate of the baseline cumulative hazard function, which facilitates the estimation of cumulative risk quantities such as survival probability contrasts that may be preferred for interpretation purposes in practice.

2.3. Inverse probability weighted score

In reality, the full data are never observed. Instead we have the observed counting process $N(t) = I(X \leq t, \Delta = 1)$ and the observed at-risk process $Y(t) = I(X \geq t)$. Define

$$M(t; \beta, \Lambda) = N(t) - \int_0^t Y(u) \exp(\beta A) \, d\Lambda(u),$$

which in general is not a martingale under model (1), posing theoretical challenges in designing an effective estimation scheme. To bridge the divide between the full data and the observed data, IPW is commonly employed. This involves weighting an observation by its inverse probability of being sampled from the target population ([Horvitz & Thompson, 1952](#)), resulting in a pseudo-random sample representing the desired population. Specifically, in the presence of nonrandomized treatment in observational studies, as well as informative censoring dependent on the covariates, [Hernán et al. \(2001\)](#) applied IPTW and IPCW to obtain consistent estimates of the parameters, provided that the relevant models are correctly specified.

Let $\pi(Z) = \text{pr}(A = 1 \mid Z)$ and $\tilde{\pi}(A, Z) = \pi(Z)^A \{1 - \pi(Z)\}^{1-A}$. In addition, let $S(t; A, Z) = \text{pr}(T > t \mid A, Z)$ and $S_c(t; A, Z) = \text{pr}(C > t \mid A, Z)$ denote the conditional survival functions of T and C , respectively. We now have the IPW scores

$$D_1^w(t; \beta, \Lambda, \pi, S_c) = \frac{dM(t; \beta, \Lambda)}{\tilde{\pi}(A, Z) S_c(t; A, Z)}, \quad (4)$$

$$D_2^w(\beta, \Lambda, \pi, S_c) = \int_0^\tau \frac{A dM(t; \beta, \Lambda)}{\tilde{\pi}(A, Z) S_c(t; A, Z)}. \quad (5)$$

It can also be readily verified that for a random sample of observed data, $\sum_{i=1}^n D_{1i}^w(t; \beta, \Lambda, \pi, S_c) = 0$ gives a weighted Breslow's estimate of $\Lambda(t)$ and, after profiling out $\Lambda(\cdot)$, $\sum_{i=1}^n D_{2i}^w(\beta, \Lambda, \pi, S_c)$ gives the weighted partial likelihood score for β . The [Supplementary Material](#) contains a formal proof of identifiability of these causal estimands via the IPW scores (4) and (5).

2.4. Augmented inverse probability weighted score

The IPW score is unbiased when the weights, or equivalently $\pi(Z)$ and $S_c(t; A, Z)$, are known ([Hernán et al., 2001](#)). In practice, these quantities are typically unknown. Propensity score models and conditional censoring models are often employed to estimate the respective conditional distributions. When these models are misspecified, the resulting estimate of the causal hazard ratio becomes inconsistent.

To protect against possible misspecification of the models, semiparametric theory has been developed to augment the IPW score ([Tsiatis, 2006](#)). The resulting AIPW score possesses the doubly robust properties that will be described in detail later. In particular, [van der Laan & Robins \(2003\)](#) augmented the IPTW score function for a binary treatment, and [Rotnitzky & Robins \(2005\)](#) augmented the IPCW score function for a survival parameter of interest. Here we apply these approaches together to the full-data martingale increments, in order to simultaneously account for confounding and informative censoring. We will show that the resulting AIPW score is doubly robust.

Denote the counting process for censoring events by $N_c(t) = I(X \leq t, \Delta = 0)$, and let $\Lambda_c(t; A, Z) = -\int_0^t S_c(u; A, Z)^{-1} dS_c(u; A, Z)$ be the cumulative hazard function of C given A and Z . Define $M_c(t; A, S_c) = N_c(t) - \int_0^t Y(u) d\Lambda_c(u; A, Z)$; then it is a martingale with respect to its natural history filtration. Following [Zhang & Schaubel \(2012b\)](#) and [Luo & Xu \(2022\)](#), define also the censor-free counting process $N_T(t) = I(T \leq t)$ and the censor-free at-risk process $Y_T(t) = I(T \geq t)$. Let $M_T(t; \beta, \Lambda) = N_T(t) - \int_0^t Y_T(u) \exp(\beta A) d\Lambda(u)$; this M_T is not a martingale in the presence of confounding.

We proceed to simultaneously augment both (4) and (5), leading to an AIPW estimate of β and Λ jointly. Joint augmentation mitigates potential bias that may arise when augmenting distinct model components separately. By considering the interdependencies among augmented elements, it enhances the precision of parameter estimates, thereby bolstering the accuracy and reliability of inferential outcomes. Our simultaneous augmentation leads to the following new estimating equations:

$$D_1(t; \beta, \Lambda, \eta) = 0, \quad D_2(\beta, \Lambda, \eta) = 0,$$

where η denotes the vector of the three nuisance functions π , S and S_c whenever possible,

$$\begin{aligned} D_1(t; \beta, \Lambda, \eta) &= \frac{dM(t; \beta, \Lambda)}{\tilde{\pi}(A, Z)S_c(t; A, Z)} - \frac{E\{dM_T(t; \beta, \Lambda) \mid A, Z\}}{\tilde{\pi}(A, Z)} \\ &\quad + \sum_{a=0}^1 E\{dM_T(t; \beta, \Lambda) \mid A = a, Z\} \\ &\quad + \sum_{a=0}^1 w^a(A, Z) \int_0^t \frac{dM_c(u; a, S_c)}{S_c(u; a, Z)} E\{dM_T(t; \beta, \Lambda) \mid T \geq u, A = a, Z\}, \end{aligned}$$

with $w^a(A, Z) = A^a(1 - A)^{1-a}\{\tilde{\pi}(A, Z)\}^{-1}$, and

$$\begin{aligned} D_2(\beta, \Lambda, \eta) &= \int_0^\tau \left[\frac{A dM(t; \beta, \Lambda)}{\pi(Z)S_c(t; A, Z)} - \frac{A E\{dM_T(t; \beta, \Lambda) \mid A, Z\}}{\pi(Z)} \right. \\ &\quad + E\{dM_T(t; \beta, \Lambda) \mid A = 1, Z\} \\ &\quad \left. + \frac{A}{\pi(Z)} \int_0^t \frac{dM_c(u; 1, S_c)}{S_c(u; 1, Z)} E\{dM_T(t; \beta, \Lambda) \mid T \geq u, A = 1, Z\} \right]. \end{aligned}$$

Simplifying the conditional expectations leads to the proposed AIPW scores

$$D_1(t; \beta, \Lambda, \eta) = d\mathcal{N}^{(0)}(t; \eta) - \Gamma^{(0)}(t; \beta, \eta) d\Lambda(t), \quad (6)$$

$$D_2(\beta, \Lambda, \eta) = \int_0^\tau d\mathcal{N}^{(1)}(t; \eta) - \Gamma^{(1)}(t; \beta, \eta) d\Lambda(t), \quad (7)$$

where for $l = 0, 1$ we have

$$\begin{aligned} d\mathcal{N}^{(l)}(t; \eta) &= \frac{A^l dN(t)}{\tilde{\pi}(A, Z)S_c(t; A, Z)} + \frac{A^l dS(t; A, Z)}{\tilde{\pi}(A, Z)} \\ &\quad - \sum_{a=0}^1 a^l \{1 + w^a(A, Z)J(t; a, S, S_c)\} dS(t; a, Z), \\ \Gamma^{(l)}(t; \beta, \eta) &= \frac{A^l Y(t) \exp(\beta A)}{\tilde{\pi}(A, Z)S_c(t; A, Z)} - \frac{A^l S(t; A, Z) \exp(\beta A)}{\tilde{\pi}(A, Z)} \\ &\quad + \sum_{a=0}^1 a^l \exp(\beta a) \{1 + w^a(A, Z)J(t; a, S, S_c)\} S(t; a, Z) \end{aligned}$$

and $J(t; a, S, S_c) = \int_0^t dM_c(u; a, S_c) / \{S(u; a, Z)S_c(u; a, Z)\}$. The first term in $d\mathcal{N}^{(l)}(t; \cdot)$ is the IPW version of $A^l dN(t)$, and the other terms constitute augmentation for both IPTW and IPCW used in the first term; these expressions are parallel to those derived in [Tsiatis \(2006\)](#) and [Bai et al. \(2017\)](#) for the simple mean of a transformed failure time. Similarly, $\Gamma^{(l)}(t; \cdot)$ can be seen as the augmented weighted $A^l Y(t) \exp(\beta A)$. The observed $A^l dN(t)$ and $A^l Y(t) \exp(\beta A)$ for $l = 0, 1$ correspond to the building blocks of the original full-data

scores in (2) and (3): $dN_T^a(t)$, $a dN_T^a(t)$, $Y_T^a(u) \exp(\beta a)$ and $a Y_T^a(u) \exp(\beta a)$. In this way the AIPW scores defined by (6) and (7) parallel the full-data scores via weighting and then augmentation.

The following theorem introduces a crucial doubly robust property inherent to the proposed population score, elucidating the significance of the new joint augmentation. This theorem represents an initial step towards the potential establishment of rate or model doubly robust properties, and to the best of our knowledge it is a novel contribution. It underscores that the population equations are centred when either one of the two key conditions holds, but not necessarily both: the first condition pertains to the accurate model specification for the conditional distribution of the failure time, while the second condition concerns the correct model specification of both missingness mechanisms, the conditional censoring and the treatment assignment probability. In the following a superscript o denotes the true value of a parameter.

THEOREM 1. *Under Assumption 1, if either $S = S^o$, or both $S_c = S_c^o$ and $\pi = \pi^o$, then for all t , $E\{D_1(t; \beta^o, \Lambda^o, \pi, S, S_c)\} = E\{D_2(\beta^o, \Lambda^o, \pi, S, S_c)\} = 0$.*

3. MISSPECIFIED COX MODEL

The proportional hazards assumption under model (1) could be violated. Without this assumption, a saturated model for the causal log hazard ratio that may change over time is

$$\lambda_{T(a)}(t) = \lambda_0(t) \exp\{\beta(t)a\}. \quad (8)$$

Obviously, $\beta(t) = \log\{\lambda_{T(1)}(t)/\lambda_{T(0)}(t)\}$. Under the saturated model (8), instead of estimating $\beta(t)$ at every t , in practice it is often of interest to estimate an average log hazard ratio,

$$\beta^* = \frac{\int_0^\tau \omega(t) \beta(t) dt}{\int_0^\tau \omega(t) dt}, \quad (9)$$

where $\omega(t) > 0$ is a weight function. If $\beta(t) = \beta_0$, then $\beta^* = \beta_0$.

Under model (8) we may still consider for the full data

$$M_T^a(t; \beta, \Lambda) = N_T^a(t) - \int_0^t Y_T^a(u) \exp(\beta a) d\Lambda(u),$$

where $\Lambda(\cdot)$ is right-continuous and nondecreasing with $\Lambda(0) = 0$ and $\Lambda(\tau) < \infty$. Observe that $M_T^a(t; \beta, \Lambda)$ is no longer a full-data martingale for any constant β , or Λ , if the proportional hazards assumption is violated, i.e., if model (1) does not hold.

To understand what the full-data scores (2) and (3) estimate under the saturated model (8), we have the following lemma.

LEMMA 1. *Under (8), the full-data equations $E\{D_1^f(t; \beta, \Lambda)\} = 0$ for each t and $E\{D_2^f(\beta, \Lambda)\} = 0$ have a unique solution $\{\beta^*, \Lambda^*(t)\}$ satisfying*

$$h(\beta^*) = \int_0^\tau \{\mathcal{E}(\beta(t), t) - \mathcal{E}(\beta^*, t)\} \sum_{a=0}^1 dF_a(t) = 0, \quad (10)$$

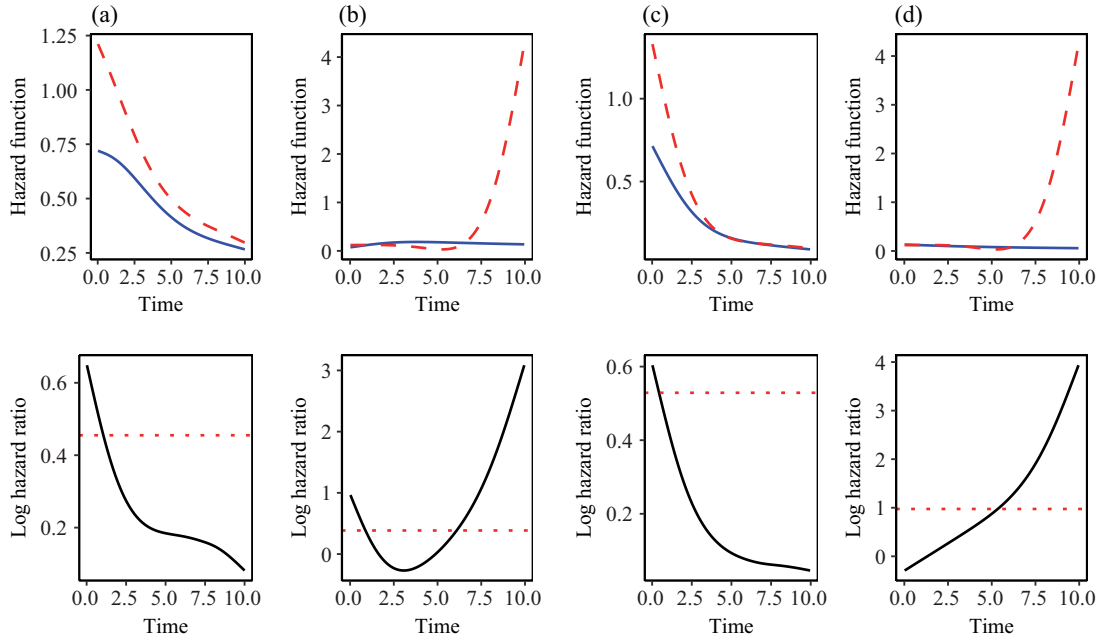


Fig. 1. Plots of the hazard functions for $a = 0$ (solid line) and $a = 1$ (dashed line) in the top row and the corresponding log hazard ratios $\beta(t)$ in the bottom row, with the dotted line in each panel indicating β^* . The following scenarios are considered: (a) $\log\{T(a)\} \sim N(-0.5a, 1)$ for $a = 0, 1$, where N denotes the normal distribution; (b) $\log\{T(0)\} \sim N(1.5, 1)$ and $T(1) \sim \text{Un}(0, 10)$; (c) $\log\{T(a)\} = -a + \epsilon$ for $a = 0, 1$ with ϵ logistic; (d) $\log\{T(0)\} = 2 + \epsilon$ with ϵ logistic, and $T(1) \sim \text{Un}(0, 10)$.

$$\Lambda^*(t) = \int_0^t \frac{\sum_{a=0}^1 dF_a(t)}{\sum_{a=0}^1 S_a(t) \exp(\beta^* a)}, \quad (11)$$

where $S_a(t) = 1 - F_a(t) = \text{pr}\{T(a) > t\}$ and $\mathcal{E}(\beta, t) = \exp(\beta)S_1(t)/\sum_{a=0}^1 \exp(\beta a)S_a(t)$.

It can be immediately verified from (10) and (11) that if $\beta(t) = \beta_0$, then $\beta^* = \beta_0$ and $\Lambda^*(\cdot) = \Lambda(\cdot)$ under the marginal structural Cox model (1). In Fig. 1 we provide some examples of $T(a)$ distributions and the corresponding β^* values. As illustrated in the figure and as is evident from Lemma 1, the estimand β^* is well specified (Buja et al., 2019b,c) as a functional of the potential outcome distributions only; it does not depend on the treatment assignment mechanism, the covariate distribution or the censoring mechanism.

To further help with interpretation, from (10), using the mean value theorem or first-order Taylor expansion we can see that (9) holds with $\omega(t) = v\{\tilde{\beta}(t), t\} \sum_{a=0}^1 f_a(t)$, where

$$v(\beta, t) = \frac{\partial}{\partial \beta} \mathcal{E}(\beta, t) = \frac{\exp(\beta)S_0(t)S_1(t)}{\{S_0(t) + \exp(\beta)S_1(t)\}^2},$$

$f_a(t) = dF_a(t)/dt$ and $\tilde{\beta}(t)$ lies between $\beta(t)$ and β^* . The above is in fact equivalent to Xu & O'Quigley (2000) under 1:1 randomized treatments, where (10) becomes

$$\int_0^\tau \{E_{\beta(t)}(A | T = t) - E_{\beta^*}(A | T = t)\} dF(t) = 0, \quad (12)$$

with $F(t)$ being the cumulative distribution function of T . In this case we also have that $\omega(t) = v\{\tilde{\beta}(t), t\}f(t)$, $v\{\beta(t), t\} = \text{var}(A \mid T = t)$, which changes very mildly over t for finite τ , and $f(t) = dF(t)/dt$.

Finally, since model (8) is saturated, we have a special case when $T(a)$ follows the semiparametric transformation model

$$g\{T(a)\} = \gamma a + \epsilon \quad (13)$$

for $a = 0, 1$, where $g(\cdot)$ is an unspecified strictly increasing function and ϵ is a member of the G^p family (Harrington & Fleming, 1982). As in Xu & Harrington (2001), it can be shown that if τ is large so that $\text{pr}\{T(a) < \tau\} = 1$ for $a = 0, 1$, then $\beta^* = -\gamma/(\rho + 1)$; see the 2023 PhD thesis by J. Luo of the University of California, San Diego. In this way estimation of the average log hazard ratio β^* leads immediately to estimation of the causal effect γ under the semiparametric transformation model (13), as $\rho \geq 0$ is assumed to be known under the model.

Once we understand what the full-data scores (2) and (3) estimate, it is perhaps not surprising that with observed data, the AIPW scores (6) and (7) identify the causal estimands β^* and Λ^* under model (8). This is shown in the following theorem, together with the doubly robust property just as previously stated under model (1), but now without the proportional hazards assumption.

THEOREM 1'. *Under Assumption 1, if either $S = S^o$, or both $S_c = S_c^o$ and $\pi = \pi^o$, then for all t , $E\{D_1(t; \beta^*, \Lambda^*, \pi, S, S_c)\} = E\{D_2(\beta^*, \Lambda^*, \pi, S, S_c)\} = 0$.*

This result stands in stark contrast to previous work under the Cox regression model, where the partial likelihood serves as the standard approach under the proportional hazards assumption, as long as the censoring distribution is independent of the failure time, given the covariates in the regression model. The marginal structural model, on the other hand, does not include any covariates, so IPCW is applied from the start, and AIPW leads to improved robustness and possibly efficiency. In this way, as will be shown in the next section, our joint augmentation provides an umbrella approach that gives valid inference for β^o under the marginal structural Cox model (1), but at the same time also valid inference for the causal estimand β^* , as well as Λ^* , under the saturated model (8). This is particularly relevant for practical applications where, as will be illustrated later, if we believe that model (1) holds, we are estimating the log hazard ratio β^o , and in the case where model (1) fails, we are estimating the time-averaged log hazard ratio β^* . The result therefore remains interpretable under a possibly misspecified marginal structural Cox model, hence satisfying both the assumption-lean and the well-specification criteria discussed in Buja et al. (2019a,b,c).

4. ASYMPTOTIC DOUBLY ROBUST PROPERTIES

4.1. Estimation

In the following we assume the general model (8), with model (1) as a special case where $\beta^* = \beta^o$. Given a random sample $(X_i, \delta_i, A_i, Z_i)$ ($i = 1, \dots, n$), we can estimate (β^*, Λ^*) by solving

$$\frac{1}{n} \sum_{i=1}^n D_{1i}(t; \beta, \Lambda, \eta) = 0, \quad \frac{1}{n} \sum_{i=1}^n D_{2i}(\beta, \Lambda, \eta) = 0. \quad (14)$$

For $l = 0, 1$, define $\mathcal{S}^{(l)}(t; \beta, \eta) = n^{-1} \sum_{i=1}^n \Gamma_i^{(l)}(t; \beta, \eta)$. Solving for Λ in (14) gives a new AIPW estimate

$$\tilde{\Lambda}(t; \beta, \eta) = \frac{1}{n} \sum_{i=1}^n \int_0^t \frac{d\mathcal{N}_i^{(0)}(u; \eta)}{\mathcal{S}^{(0)}(u; \beta, \eta)}. \quad (15)$$

With it, we can go back to (14) and obtain a new AIPW estimate of β that solves the estimating equation $U(\beta; \eta) = 0$, where

$$U(\beta; \eta) = \frac{1}{n} \sum_{i=1}^n \int_0^\tau d\mathcal{N}_i^{(1)}(t; \eta) - \bar{A}(t; \beta, \eta) d\mathcal{N}_i^{(0)}(t; \eta) \quad (16)$$

with $\bar{A}(t; \beta, \eta) = \mathcal{S}^{(1)}(t; \beta, \eta)/\mathcal{S}^{(0)}(t; \beta, \eta)$. We remark that $U(\beta; \eta)$ has a parallel expression to the partial likelihood score function under the Cox model, upon observing that $\bar{A}(t; \cdot)$ is the sum of the augmented $A_i \exp(\beta A_i) Y_i(t)$ divided by the sum of the augmented $\exp(\beta A_i) Y_i(t)$, and $d\mathcal{N}_i^{(l)}(t; \eta)$ is the augmented $A_i^l dN_i(t)$ for $l = 0, 1$ as mentioned earlier. As demonstrated below, these augmentations yield the doubly robust property for inference in large samples. The above derivation and result are, to the best of our knowledge, new to the literature.

The nuisance functions $\pi(z)$, $S(t; a, z)$ and $S_c(t; a, z)$ are unknown in practice and need to be estimated. We employ the cross-fitting procedure that is commonly used in the literature (Hasminskii & Ibragimov, 1979; Bickel, 1982; Robins et al., 2008; Chernozhukov et al., 2018) and, as will be shown later, provides root- n inference for β^* . The estimation procedure is summarized in Algorithm 1, where the details of the relevant quantities are given in the Supplementary Material, though the notation used here should be self-explanatory. Following the fit, under the marginal structural Cox model (1), if needed we may also estimate the survival probabilities, or 1 minus the risk, $\text{pr}\{T(a) > t\}$, by $\exp\{-\hat{\Lambda}(t) \exp(\hat{\beta}a)\}$.

Algorithm 1. Jointly augmented Cox doubly robust estimator.

Require: a sample of n observations, $(X_i, \delta_i, A_i, Z_i)_{i=1}^n$.

Split the full sample into k folds indexed by $\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_k$

for each fold indexed by m do.

Let $\mathcal{I}_{-m} = \{1, \dots, n\} \setminus \mathcal{I}_m$.

Estimate nuisances $\hat{\eta}^{(-m)} = (\hat{\pi}^{(-m)}, \hat{S}^{(-m)}, \hat{S}_c^{(-m)})$
using the out-of-fold sample $\mathcal{I}_{-m}$.

Obtain $U_m(\beta, \hat{\eta}^{(-m)})$, the m th-fold estimating equation for β ,
by profiling out Λ from the in-fold equation $\sum_{i \in \mathcal{I}_m} D_{1i}(t; \beta, \Lambda, \hat{\eta}^{(-m)}) = 0$.

Construct $\tilde{\Lambda}_m(t; \hat{\beta}, \hat{\eta}^{(-m)})$ as in (15).

end for

Return $\hat{\beta}$ as the solution to $U_{cf}(\beta) = 0$ with

$$U_{cf}(\beta) = \frac{1}{k} \sum_{m=1}^k U_m(\beta, \hat{\eta}^{(-m)}), \quad \hat{\Lambda}(t) = \frac{1}{k} \sum_{m=1}^k \tilde{\Lambda}_m(t; \hat{\beta}, \hat{\eta}^{(-m)}).$$

4.2. Asymptotic properties and inference

We first focus on rate double robustness and describe some additional assumptions. Let $O^\dagger$ denote a random sample $\{(X_i^\dagger, \Delta_i^\dagger, A_i^\dagger, Z_i^\dagger), i = 1, \dots, n\}$ used for estimating $\hat{\pi}$, $\hat{S}$ and $\hat{S}_c$. Let (X, Δ, A, Z) be a data point independent of $O^\dagger$ and drawn from the same distribution as $O^\dagger$. For $f = S, S_c$ define

$$\begin{aligned}\|\hat{\pi} - \pi^*\|_{\dagger}^2 &= E^\dagger[E\{\hat{\pi}(Z) - \pi^*(Z)\}^2], \\ \|\hat{f} - f^*\|_{\dagger}^2 &= E^\dagger\left[E\left\{\sup_{t \in [0, \tau], a \in \{0, 1\}} |\hat{f}(t; a, Z) - f^*(t; a, Z)|^2\right\}\right],\end{aligned}$$

where $E^\dagger$ denotes expectation with respect to $(X^\dagger, \Delta^\dagger, A^\dagger, Z^\dagger)$ and E denotes expectation with respect to (X, Δ, A, Z) conditional on $O^\dagger$.

Assumption 2 (Uniform convergence). There exist deterministic limits $\pi^*(z)$, $S^*(t; a, z)$ and $S_c^*(t; a, z)$ such that $\|\hat{\pi} - \pi^*\|_{\dagger} = o(1)$, $\|\hat{S} - S^*\|_{\dagger} = o(1)$ and $\|\hat{S}_c - S_c^*\|_{\dagger} = o(1)$.

This assumption does not require any of the three models to be correctly specified. Rather, it simply asserts that these models possess population counterparts to which they converge. This condition is generally satisfied for the Donsker model class, regardless of whether the models exhibit biases. The following theorem establishes doubly robust consistency of $\hat{\beta}$ in cases where either the outcome model, i.e., the failure time model, or the missingness model, i.e., the censoring time model and the treatment assignment model, is correctly specified, but not necessarily both, and our proposed estimator remains valid.

THEOREM 2 (CONSISTENCY). *Under Assumptions 1 and 2 and additional regularity assumptions given in the [Supplementary Material](#), if either $S^* = S^o$ or $(\pi^*, S_c^*) = (\pi^o, S_c^o)$, then $\hat{\beta} \rightarrow \beta^*$ in probability.*

As a reminder, the limit β^* to which $\hat{\beta}$ converges is the true log hazard ratio when the marginal structural Cox model (1) holds; otherwise it is the time-averaged log hazard ratio described in §3.

Now let O denote a random sample $\{(X_j, \Delta_j, A_j, Z_j), j = 1, \dots, n\}$ that is independent of $O^\dagger$ and drawn from the same distribution as $O^\dagger$. Recall that $\eta^o = (\pi^o, S^o, S_c^o)^\top$, and define $\tilde{J}_i(t; a) = J_i(t; a, \hat{S}, \hat{S}_c) - J_i(t; a, \hat{S}, S_c^o)$ and $\bar{J}_i(t; a) = J_i(t; a, S^o, \hat{S}_c) - J_i(t; a, S^o, S_c^o)$, with $J_i(t; a, S, S_c) = \int_0^t dM_{ci}(u; a, S_c) / \{S_i(u; a)S_{ci}(u; a)\}$ as defined in §2. Here we use shorthand notation such as $\hat{S}_i(t; a) = \hat{S}(t; a, Z_i)$ and $S_i^o(t; a) = S^o(t; a, Z_i)$. Then, with $d\mathcal{K}_i(t, u; a) = d\hat{S}_i(t; a) d\tilde{J}_i(u; a) - dS_i^o(t; a) d\bar{J}_i(u; a)$, we define the cross integral products $\mathcal{D}_1^\dagger = \mathcal{D}_1^\dagger(\hat{S}, \hat{S}_c; \eta^o)$ and $\mathcal{D}_2^\dagger = \mathcal{D}_2^\dagger(\hat{S}, \hat{S}_c; \eta^o)$ as

$$\begin{aligned}\mathcal{D}_1^\dagger &= E^\dagger\left(E\left[\max_{a \in \{0, 1\}} \frac{1}{n} \sum_{i=1}^n \left| \int_0^\tau \int_0^t \{a - \bar{A}(t; \beta^*, \eta^o)\} d\mathcal{K}_i(t, u; a) \right| \right]\right), \\ \mathcal{D}_2^\dagger &= E^\dagger\left(E\left[\max_{a, l \in \{0, 1\}} \frac{1}{n} \sum_{i=1}^n \left| \int_0^\tau \delta(t, \beta^*) J_i(t; a, S^o, \hat{S}_c)^l \{d\hat{S}_i(t; a) - dS_i^o(t; a)\} \right| \right]\right),\end{aligned}$$

where $\delta(t, \beta^*) = \bar{A}(t; \beta^*, \pi^o, S^o, \hat{S}_c) - \bar{A}(t; \beta^*, \eta^o)$ and E denotes expectation with respect to O conditional on $O^\dagger$.

Assumption 3 (Rate condition). We have that

$$\|\hat{S} - S^o\|_{\dagger}(\|\hat{\pi} - \pi^o\|_{\dagger} + \|\hat{S}_c - S_c^o\|_{\dagger}) + \mathcal{D}_1^\dagger(\hat{S}, \hat{S}_c; \eta^o) + \mathcal{D}_2^\dagger(\hat{S}, \hat{S}_c; \eta^o) = o(n^{-1/2}).$$

In addition to the common products of error rates as in Chernozhukov et al. (2018), Rotnitzky et al. (2021) and Hou et al. (2023), we also have the cross integral product terms. These integral terms are needed because we have two nuisance functions that involve time t ; in contrast, the mixed bias property of Rotnitzky et al. (2021) suffices when at most one of the nuisance functions involves t , as in their cases. Similar integral terms in rate conditions can be found in Wang et al. (2024) and Vansteelandt et al. (2024).

THEOREM 3 (RATE DOUBLE ROBUSTNESS). *Under Assumptions 1–3 and additional regularity assumptions given in the Supplementary Material, if $(\pi^*, S^*, S_c^*) = (\pi^o, S^o, S_c^o)$, then*

$$\hat{\sigma}^{-1} n^{1/2}(\hat{\beta} - \beta^*) \rightarrow N(0, 1)$$

in distribution, where the expression for $\hat{\sigma}^2 = \hat{\sigma}^2(\hat{\beta})$ is provided in the Supplementary Material.

Theorem 3 establishes the rate double robustness property, and as mentioned before this solves the compatibility issue due to the non-collapsibility of the Cox regression model via the use of machine learning or other nonparametric methods to estimate the nuisance functions. Traditionally, Neyman orthogonal scores (Neyman, 1959) have been considered in the semiparametric literature, which, when combined with cross-fitting, yield $\sqrt{n}$ -consistent estimators, as long as all nuisance parameters are estimated at faster than $n^{-1/4}$ rate (Newey, 1994; Rotnitzky et al., 2021). However, this $n^{-1/4}$ rate requirement still rules out a number of data-adaptive machine learning methods (Bilodeau, 2022; Ogburn et al., 2022; Tang, 2022). The rate double robustness result here improves upon the $n^{-1/4}$ rate requirement: $\hat{\beta}$ is consistent and asymptotically normal even if one of $\hat{S}$ or $(\hat{\pi}, \hat{S}_c)$ converges arbitrarily slow, as long as their product error rate is faster than $\sqrt{n}$. In practice, very few of the machine learning methods used for time-to-event data have known convergence rates, and in §5 we will investigate empirically the performance of the methods.

Lastly, if we are satisfied with the saturated model (8) and do not insist on model (1) being valid, we can explore parametric or semiparametric models for the nuisance functions π , S and S_c , as the compatibility issue is no longer a concern. In this case, no cross-fitting is needed, and $\hat{\pi}$, $\hat{S}$ and $\hat{S}_c$ are regular and asymptotically linear estimators of π , S and S_c , respectively. Let π^* , S^* and S_c^* be their limits; that is, $\|\hat{\pi} - \pi^*\| = \|\hat{S} - S^*\| = \|\hat{S}_c - S_c^*\| = o(1)$, where $\|\hat{\pi} - \pi^*\|^2 = E\{|\hat{\pi}(Z) - \pi^*(Z)|^2\}$, $\|\hat{S} - S^*\|^2 = E\{\sup_{t \in [0, \tau], a \in \{0, 1\}} |\hat{S}(t; a, Z) - S^*(t; a, Z)|^2\}$ and $\|\hat{S}_c - S_c^*\|^2 = E\{\sup_{t \in [0, \tau], a \in \{0, 1\}} |\hat{S}_c(t; a, Z) - S_c^*(t; a, Z)|^2\}$. As in Hou et al. (2023) and Wang et al. (2024), it can be shown that the solution $\hat{\beta}_{dr}$ to $U(\beta; \hat{\eta}) = 0$ in (16) enjoys the classical model doubly robust property: it is consistent and asymptotically normal if either $S^* = S^o$ or $(\pi^*, S_c^*) = (\pi^o, S_c^o)$. In addition, since (8) is in fact a nonparametric model, the influence function for β^* is unique and hence efficient (Tsiatis, 2006); that is, $\hat{\beta}$ attains the semiparametric efficiency bound if all the nuisance functions are correctly estimated.

Table 1. Four scenarios for generating $C(a)$ and A

Scenario	Details
1 $C(a)$: Cox A : Logistic	$\epsilon \sim \text{Un}(0, 1)$ $C(a) = -\log(\epsilon) \exp(0.5 + 0.5a - Z_2 + 0.5Z_3)$ $\text{logit}\{\pi(Z)\} = 0.5Z_1 - 0.5Z_2 - 0.5Z_3$
2 $C(a)$: Cox A : Soft partition	$\epsilon \sim \text{Un}(0, 1)$ $C(a) = -\log(\epsilon) \exp(0.5 + 0.5a - Z_2 + 0.5Z_3)$ $\text{logit}\{\pi(Z)\} = -3I\{Z_2 < -0.5\} + 3I\{-0.5 \leq Z_2 < 0.5\} - 3I\{Z_2 \geq 0.5\}$
3 $C(a)$: Uniform-Cox A : Logistic	$\epsilon \sim \text{Un}(0, 1)$ $C(0) = 1.05\epsilon, C(1) = -\log(\epsilon) \exp(3.3 + 3.5Z_3)$ $\text{logit}\{\pi(Z)\} = 0.5Z_1 - 0.5Z_2 - 0.5Z_3$
4 $C(a)$: Uniform-Cox A : Soft partition	$\epsilon \sim \text{Un}(0, 1)$ $C(0) = 1.05\epsilon, C(1) = -\log(\epsilon) \exp(3.3 + 3.5Z_3)$ $\text{logit}\{\pi(Z)\} = -3I\{Z_2 < -0.5\} + 3I\{-0.5 \leq Z_2 < 0.5\} - 3I\{Z_2 \geq 0.5\}$

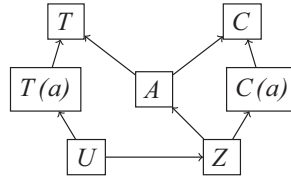


Fig. 2. Diagram for the data-generating mechanism.

5. NUMERICAL WORK

5.1. Simulations

In this section, we assess the performance of the proposed estimator under the marginal structural Cox model (1). Extended simulations under the general model (8) are reported in the [Supplementary Material](#).

To induce confounding within a marginal structural model, we adopt the approach outlined by [Hovercraft & Didelez \(2012\)](#). This involves simulating latent variables linked to both the covariates and the event outcome. Let $U = (U_1, U_2, U_3)^T$, where $U_1 \sim \text{Un}(-1, 1)$, $U_2 \sim \text{Un}(-1, 1)$ and $U_3 \sim \text{Un}(-1, 1)$. Let $Z = (Z_1, Z_2, Z_3)^T$, where $Z_1 = 0.5U_1 + U_3$, $Z_2 = U_1 + 1.5U_1^2 - 0.5$ and $Z_3 = U_1 + U_2$. Let $T(a) = -\log(0.5U_1 + 0.5) \exp(a)$ for $a = 0, 1$. This gives $T(a)$ which follows model (1) with $\beta^0 = -1$ and $\lambda_0^0(t) = 1$. We then simulate $C(a)$ and A according to the four scenarios described in Table 1, each giving about 50% per treatment arm and about 40% censored due to loss to follow-up. Finally, let $T = AT(1) + (1 - A)T(0)$ and $C = AC(1) + (1 - A)C(0)$. For all simulations we set $\tau = 1$; this is the maximum time for estimating both the T and the C distributions. The details of this data-generation process are shown in Fig. 2. Each scenario involves 1000 simulated datasets of $n = 1000$, yielding a margin of error of about $\pm 1.35\%$ for the 95% confidence intervals.

We compare the AIPW estimator, using different working models, with the IPW estimator, the partial likelihood estimator under the naive Cox model and a full-data estimator. We estimate the conditional distributions of T and C given A and Z using Cox

Table 2. *Simulation results in Scenarios 1 and 2, with true $\beta^0 = -1$; italics indicate that the model is wrong*

Scenario	Estimator	<i>T/C-A models</i>	Bias	SD	SE model/bootstrap	Coverage model/bootstrap
1	AIPW	<i>Cox/Cox-logit</i>	0.002	0.059	0.060/0.059	0.95/0.94
		<i>Cox/Cox-GBM</i>	0.003	0.058	0.061/0.060	0.96/0.94
		<i>Cox/RSF-logit</i>	0.004	0.057	0.060/0.059	0.96/0.96
		<i>Cox/RSF-GBM</i>	0.003	0.056	0.062/0.060	0.97/0.96
		<i>RSF/Cox-logit</i>	0.002	0.053	0.053/0.053	0.95/0.94
		<i>RSF/Cox-GBM</i>	0.008	0.054	0.054/0.055	0.95/0.95
		<i>RSF/RSF-logit</i>	0.005	0.053	0.053/0.054	0.95/0.94
		<i>RSF/RSF-GBM</i>	0.011	0.054	0.055/0.056	0.95/0.95
	IPW	<i>Cox-logit</i>	0.000	0.067	— /0.066	— /0.94
		<i>Cox-GBM</i>	0.031	0.069	— /0.069	— /0.92
		<i>RSF-logit</i>	0.026	0.060	— /0.064	— /0.95
		<i>RSF-GBM</i>	0.005	0.062	— /0.068	— /0.97
	<i>Naïve Cox</i>		0.496	0.100	0.100/0.101	0.00/0.00
	<i>Full-data</i>		0.001	0.029	0.028/ —	0.95/ —
2	AIPW	<i>Cox/Cox-logit</i>	0.260	0.064	0.068/0.066	0.02/0.02
		<i>Cox/Cox-GBM</i>	0.018	0.086	0.093/0.088	0.96/0.95
		<i>Cox/RSF-logit</i>	0.268	0.063	0.069/0.066	0.02/0.02
		<i>Cox/RSF-GBM</i>	0.033	0.080	0.090/0.084	0.95/0.94
		<i>RSF/Cox-logit</i>	0.050	0.071	0.056/0.061	0.80/0.83
		<i>RSF/Cox-GBM</i>	0.003	0.073	0.073/0.075	0.94/0.94
		<i>RSF/RSF-logit</i>	0.052	0.071	0.056/0.061	0.80/0.83
		<i>RSF/RSF-GBM</i>	0.009	0.073	0.073/0.075	0.95/0.94
	IPW	<i>Cox-logit</i>	0.164	0.093	— /0.092	— /0.58
		<i>Cox-GBM</i>	0.103	0.118	— /0.106	— /0.83
		<i>RSF-logit</i>	0.177	0.084	— /0.088	— /0.48
		<i>RSF-GBM</i>	0.134	0.107	— /0.101	— /0.74
	<i>Naïve Cox</i>		0.579	0.119	0.125/0.125	0.00/0.00
	<i>Full-data</i>		0.001	0.029	0.028/ —	0.95/ —

SD, standard deviation; SE, standard error; GBM, gradient boosted model; RSF, random survival forest.

regression and the random survival forest (RSF) (Ishwaran et al., 2008), from the R (R Development Core Team, 2025) package randomForestSRC. For A given Z , we used logistic regression and the gradient boosted model (GBM), from the R package twang (Ridgeway et al., 2022). RSF settings included the `bs.gradient` split rule, while default settings were maintained for other hyperparameters in both twang and RSF. Five-fold cross-fitting was implemented. For stability, values of $\hat{S}(t; a, z)$ and $\hat{S}_c(t; a, z)$ were restricted to be at least 0.05, and $\hat{\pi}(z)$ was between 0.1 and 0.9. For each dataset, we calculated model-based and bootstrap standard errors, the latter with 100 replicates, and constructed 95% confidence intervals using the normal approximation. For the full-data estimator, the robust sandwich estimator was applied to account for potential intra-subject outcome correlations.

Tables 2 and 3 show the bias, standard deviation, standard error and coverage probabilities of the estimators in Scenarios 1–4. Additional plots for visualizing the results are provided in the [Supplementary Material](#). We see that under Scenario 1 all eight AIPW

Table 3. Simulation results in Scenarios 3 and 4, with true $\beta^0 = -1$; italics indicate that the model is wrong

Scenario	Estimator	<i>T/C-A Models</i>	Bias	SD	SE model/bootstrap	Coverage model/bootstrap
3	AIPW	<i>Cox/Cox-logit</i>	0.146	0.106	0.108/0.109	0.79/0.78
		<i>Cox/Cox-GBM</i>	0.146	0.108	0.114/0.120	0.81/0.83
		<i>Cox/RSF-logit</i>	0.015	0.103	0.122/0.118	0.98/0.97
		<i>Cox/RSF-GBM</i>	0.014	0.104	0.126/0.125	0.98/0.98
		<i>RSF/Cox-logit</i>	0.007	0.092	0.098/0.097	0.95/0.94
		<i>RSF/Cox-GBM</i>	0.001	0.095	0.101/0.104	0.95/0.95
		<i>RSF/RSF-logit</i>	0.019	0.097	0.102/0.104	0.96/0.95
		<i>RSF/RSF-GBM</i>	0.026	0.098	0.103/0.111	0.96/0.97
	IPW	<i>Cox-logit</i>	0.305	0.112	— /0.103	— /0.21
		<i>Cox-GBM</i>	0.335	0.115	— /0.104	— /0.16
		<i>RSF-logit</i>	0.077	0.078	— /0.079	— /0.84
		<i>RSF-GBM</i>	0.109	0.082	— /0.082	— /0.73
	<i>Naïve Cox</i>		0.895	0.108	0.110/0.110	0.00/0.00
	<i>Full-data</i>		0.001	0.029	0.028/ —	0.95/ —
4	AIPW	<i>Cox/Cox-logit</i>	0.347	0.120	0.138/0.143	0.14/0.24
		<i>Cox/Cox-GBM</i>	0.135	0.131	0.116/0.129	0.72/0.78
		<i>Cox/RSF-logit</i>	0.361	0.201	0.343/0.224	0.50/0.71
		<i>Cox/RSF-GBM</i>	0.084	0.140	0.149/0.157	0.88/0.90
		<i>RSF/Cox-logit</i>	0.075	0.107	0.101/0.105	0.89/0.89
		<i>RSF/Cox-GBM</i>	0.026	0.105	0.096/0.106	0.90/0.92
		<i>RSF/RSF-logit</i>	0.143	0.137	0.174/0.155	0.88/0.93
		<i>RSF/RSF-GBM</i>	0.020	0.124	0.119/0.132	0.94/0.94
	IPW	<i>Cox-logit</i>	0.063	0.130	— /0.124	— /0.90
		<i>Cox-GBM</i>	0.249	0.130	— /0.117	— /0.41
		<i>RSF-logit</i>	0.163	0.112	— /0.104	— /0.66
		<i>RSF-GBM</i>	0.005	0.129	— /0.112	— /0.91
	<i>Naïve Cox</i>		0.562	0.120	0.132/0.132	0.00/0.00
	<i>Full-data</i>		0.001	0.028	0.028/ —	0.95/ —

estimators have small biases and good coverage probabilities, even when the conditional Cox model for T is wrong (marked in italics). This is no longer the case under Scenarios 2 and 3, when both the conditional Cox model for T and one of the models for C or A is wrong. Under Scenario 2, when RSF is used for the conditional T model and logistic regression is incorrectly applied for the propensity score model, the biases are relatively substantial, with coverage at only around 80% in both cases. We believe that this has to do with the fact that rate double robustness when machine learning is involved requires all nuisance estimates to be consistent, at least theoretically. Under Scenario 4 the RSF/RSF-GBM estimator using all machine learning methods has the smallest bias and is the only one with correct coverage. The IPW estimators perform well under Scenario 1 with correctly specified working models for C and A . However, their performance is generally poor in Scenarios 2–4, including the RSF-GBM estimator, which exhibits slower convergence when machine learning methods are used for weight estimation.

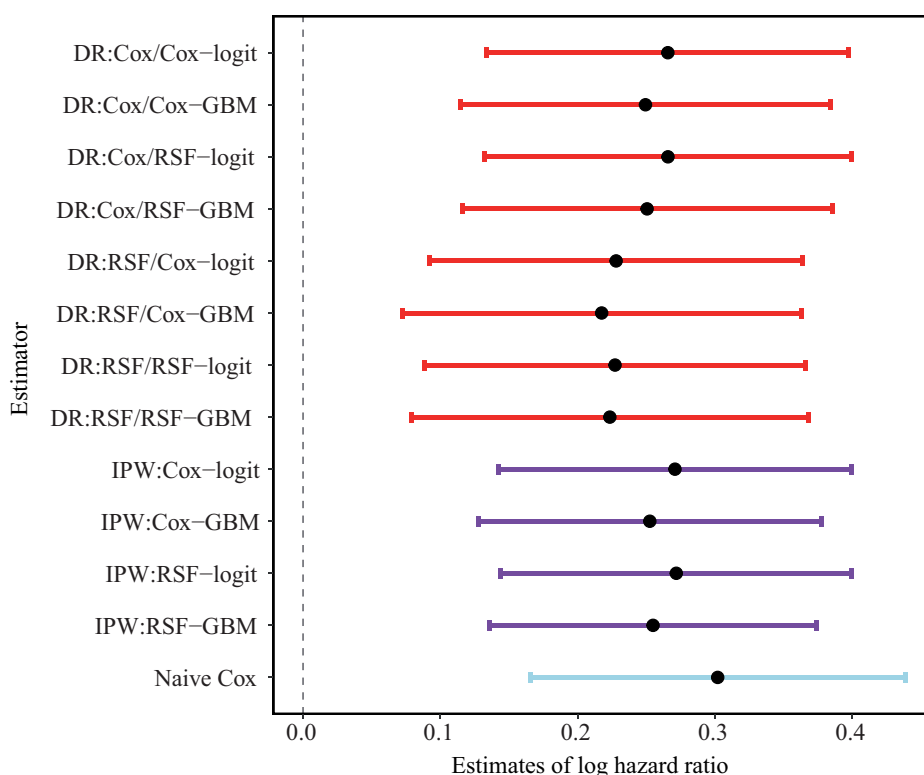


Fig. 3. Forest plot of the (time-averaged) log hazard ratio estimates of the effect of mid-life alcohol exposure (heavy versus not heavy) on overall survival for the HAAS data.

5.2. Honolulu-Asia Aging Study

We analyse data from the Honolulu-Asia Aging Study (HAAS), which began in 1991 as an extension of the Honolulu Heart Program project. The objective is to examine the impact of various mid- and late-life exposures on outcomes including cognitive and motor impairment, stroke, other common chronic conditions and mortality. Both studies followed a cohort of Japanese men born between 1900 and 1919 who were residing in Hawaii. The dataset comprises 2079 participants, with 552 classified as heavy drinkers (more than two drinks per day) at some point during mid-life. We consider covariates such as age, systolic blood pressure, heart rate at baseline, and years of education attained. The maximum follow-up duration is set at 13 years. In total, 47% of participants died, 21% were censored due to loss to follow-up, and the remaining 32% were alive at the 13-year mark. The specific focus here is the impact of mid-life alcohol exposure on late-life mortality.

Figure 3 shows the point estimates and the corresponding 95% bootstrap-based confidence intervals for all 13 estimators; a corresponding table is provided in the [Supplementary Material](#). The naive Cox estimate yields the largest point estimate at 0.30, whereas the IPW estimates are smaller after adjusting for confounding and informative censoring. Notably, the AIPW estimates, particularly those employing machine learning methods, show considerably smaller log hazard ratio estimates than the IPW estimates. All estimates confirm a statistically significant increased risk of mortality associated with mid-life alcohol exposure, although the precision of these estimates varies. The lack of validity of the IPW-based inference renders the confidence intervals questionable. Additionally, our approach

provides estimates of the risk difference $\text{pr}\{T(1) \leq t\} - \text{pr}\{T(0) \leq t\}$ and the risk ratio $\text{pr}\{T(1) \leq t\}/\text{pr}\{T(0) \leq t\}$; see the [Supplementary Material](#) for a figure showing results for $t = 3, 4, \dots, 12$ years.

When we refer to the log hazard ratio, we assume that the proportional hazards assumption holds. To date, there is a notable lack of diagnostic tools designed for assessing the marginal structural Cox model. Nevertheless, as discussed in the next section, the development of such tools appears feasible with the aid of the AIPW estimators introduced in this paper, which would hold significant practical significance.

If there are concerns about the proportional hazards assumption, interpreting the estimand as the time-averaged log hazard ratio is a useful approach. This interpretation is always useful, as discussed in [Buja et al. \(2019b\)](#), where it is noted that testing the assumption on the same dataset can lead to post-selection inference issues.

6. SPECIAL CASES AND EXTENSIONS

Our development has two special cases. The first case concerns causal inference with non-randomized observational data, but no informative censoring; in this case, only AIPTW is needed. The second case concerns informative censoring only, without the necessity to account for confounding. This may be for a treatment effect under randomization, or under a general regression setting for the observed outcomes instead of the potential outcomes; in this case, only AIPCW is needed. Both these cases, as well as the general development concerning both causal inference with observational data and informative censoring as described in this paper, have been implemented in the R package `CoxAIPW`, which is available on CRAN. The package allows three choices corresponding to the above.

While the partial likelihood might be much better known under the Cox model, the analogues of the full-data estimating equations (2) and (3) using martingale-type increments were historically developed for other semiparametric survival models, such as the semiparametric additive hazards model ([Lin & Ying, 1994](#)), the transformation model ([Chen et al., 2002](#)) and the transformation cure model ([Lu & Ying, 2004](#)). They are, in fact, much more broadly applicable to other semiparametric survival models. Thus these models may be used for potential failure time outcomes, and the joint augmentation technique developed here may be similarly applied, so that doubly robust approaches can be developed for other marginal structural models besides the Cox one, for analysing observational studies with survival outcomes.

Here we illustrate the approach of applying joint augmentation to the marginal structural transformation model (13), where γ is the treatment effect and g the infinite-dimensional nuisance parameter. We do not assume that the error comes from the G^p family, and we let Λ_ϵ denote the known cumulative distribution function of ϵ . It follows from [Chen et al. \(2002\)](#) that

$$M_T^a(t; \gamma, g) = N_T^a(t) - \int_0^t Y_T^a(u) d\Lambda_\epsilon\{\gamma a + g(u)\}$$

is a full-data martingale with respect to the filtration $\mathcal{F}_t^a = \{N_T^a(u), Y_T^a(u^+) : 0 \leq u \leq t\}$. As in §2, we have the full-data scores for γ and g ,

$$D_1^f(t; \gamma, g) = \sum_{a=0}^1 dM_T^a(t; \gamma, g),$$

$$D_2^f(\gamma, g) = \sum_{a=0}^1 \int_0^\tau a dM_T^a(t; \gamma, g).$$

Next, we can construct IPW scores $D_1^w(t; \gamma, g, \pi, S_c)$ and $D_2^w(\gamma, g, \pi, S_c)$, which take the same form as the IPW scores in §2 if we replace $M(t; \beta, \Lambda)$ with $M(t; \gamma, g) = N(t) - \int_0^t Y(u) d\Lambda_\epsilon\{\gamma A + g(u)\}$. Applying simultaneous augmentation leads to our candidate doubly robust scores $D_1(t; \gamma, g, \eta)$ and $D_2(\gamma, g, \eta)$, which also take the same form as in §2 if we further replace $M_T(t; \beta, \Lambda)$ with $M_T(t; \gamma, g) = N_T(t) - \int_0^t Y_T(u) d\Lambda_\epsilon\{\gamma A + g(u)\}$.

Martingales have also been defined for complex failure time outcomes such as semi-competing risks (Andersen et al., 1991), and we recently proposed a marginal structural illness-death model for causal inference using IPW (Zhang et al., 2024). Work is currently in progress to develop an AIPW approach using the joint augmentation technique for semi-competing risks. Finally, we have also developed an AIPW approach under covariate-induced dependent left truncation (Wang et al., 2024), and work is under way to extend it to semiparametric models using joint augmentation.

7. DISCUSSION

We have derived an AIPW estimator for the log hazard ratio under the marginal structural Cox model, both with the proportional hazard assumption and allowing it to be violated. The estimator possesses the rate doubly robust property, which provides a solution to the challenge otherwise posed by the non-collapsibility of the Cox model. In this way it becomes possible to protect against misspecification of the propensity score model, which would otherwise invalidate the IPW approach. We believe that in place of the IPW estimator commonly employed in practice, the AIPW estimator should be routinely used. In addition, as indicated in J. Luo's 2023 PhD thesis, the explicit expression of the AIPW estimator can potentially lead to diagnostics of the proportional hazard assumption in the marginal structural Cox model, and this would be of interest for future work.

In the context of model misspecification, we could not agree more with Cox (1995) that models are simplifications and idealizations of reality. While it is important to carefully examine any estimation procedure under the model assumptions, it is also very useful to know, in the often inevitable situations where a model is misspecified, what we are estimating. While Buja et al. (2019a,c) did not consider censored survival data in particular, in their rejoinder Buja et al. (2019b, p. 617) emphasized the importance for a causal estimand to not depend on the covariate distribution, and that would be 'the definition of well-specification' of such an estimand. Our development in this paper has clearly established that the estimands β^* and Λ^* , outside of the marginal structural Cox model, are functionals of the potential outcome distributions only and do not depend on either the treatment assignment mechanism, the covariate distribution or the censoring mechanism. Thus they remain interpretable in the absence of randomization, as well as not affected by specific recruitment patterns of any studies which translate to censoring mechanisms. We therefore believe that they meet the criteria of well-specification, albeit possibly in an extended sense of the original definition (Buja et al., 2019a,c).

In the discussion surrounding Buja et al. (2019a,c), Whitney et al. (2019) advocated for an estimand closely related to ours: $\int \beta(t)f(t) dt$. It is also possible to incorporate different user-defined weights into the average over time, and Whitney et al. (2019) advocated for a model-free or model-agnostic approach, which seems natural in that setting. In contrast, our work starts with the marginal structural Cox model and is closer in spirit to Buja et al. (2019c) in that we have leveraged the model to suggest the estimating functions for the larger nonparametric class of the joint distribution. Vansteelandt et al. (2024) also focused on assumption-lean Cox regression. They considered an averaged log cumulative hazard ratio as the estimand, but the time interval on which the average is taken has to be bounded away from time zero. On the other hand, the proportional hazards assumption, if it holds at all, is more likely to be satisfied before too much time has elapsed. In this sense, the approach seems more like an alternative rather than assumption-lean.

A separate line of work by Lu & Tsiatis (2008) considered efficient testing under randomization, making use of auxiliary covariate information. Their approach used modern semiparametric theory that had led to the development of AIPTW and AIPCW in the literature (Tsiatis, 2006). An advantage of a log-rank-type test is that it is in fact model-free, so that its power can remain good under certain model misspecifications. We point out the difference in assumptions about censoring: in order to be comparable to the log-rank test in the two-group setting, Lu & Tsiatis (2008) required censoring to be independent of the failure time and the covariates given the group assignments.

Our AIPW approach with the saturated model may also offer a novel perspective on handling situations with a large number of covariates or effect modifiers, addressing both precise parameter identification and high dimensionality. The concept of an average, also referred to as a least-true, parameter introduces a fresh viewpoint with the potential for more efficient utilization of high-dimensional structures than precise models. While significant and promising, this extension falls beyond the scope of the present study; nevertheless, our joint augmentation framework has laid the groundwork for future exploration.

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SUPPLEMENTARY MATERIAL

The [Supplementary Material](#) contains notation and expressions, some results under misspecified Cox models, proofs of identifiability and double robustness, additional plots, simulation results under the $\beta(t)$ model and proofs of the asymptotic results.

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